

Scalability & Future Plan

Date	02 October 2026
Team ID	6ab3ed02b5170a9b58b3aa3a
Project Name	ComicCraft AI
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Local file storage	Harder to scale for many users	High
2	External AI API limits	Generation may be limited by provider quotas	High
3	Single-instance deployment	Concurrent users may increase response time	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Render service	Use scalable cloud instances and request queuing
2	Data Storage	Local generated files	Move assets to cloud object storage
3	Performance	Synchronous generation flow	Add background jobs, caching and progress tracking
4	Security	Environment variables for API keys	Add authentication, access control and secret management

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Improve UI and error handling	1-2 months	Better usability
Phase 3	Cloud storage and background jobs	2-4 months	Higher scalability
Phase 4	User accounts and comic library	4-6 months	Personalized usage